DTSC 2302 - Project 1 Part 1 - Group: Pavle D, Ikumi U, Sonia S, Michael F, and James H

Student Academic Performance Analysis and Predicting Final Grades Using OLS Regression

Model Building Process

The purpose of this model is to predict students' final grades (G3) using all available predictors in the dataset. The data was first checked for any unnecessary columns, and the extra index column was removed. Since several variables were categorical, they were converted into dummy variables using one-hot encoding with the drop-first method so they could be properly included in the regression model while avoiding redundancy. The target variable, G3, was then separated from the predictors, and a constant term was added to meet the requirements of the OLS model.

To evaluate how well the model performs on new data and reduce the risk of overfitting, the dataset was split into an 80% training set and a 20% testing set. The model was fit using the training data, and its performance was then evaluated using both in-sample and out-of-sample metrics. Although the original model included all predictors, feature selection was also explored using backward elimination at a significance level of 0.05 on the training data. This allowed us to evaluate whether a more concise model could maintain similar predictive performance while improving interpretability. All in all, the final model kept all predictors for comparison purposes, as key variables remained statistically significant and contributed to overall model performance.

Prediction Model and Coefficient Interpretation

The final model is a multiple linear regression that estimates G3 as a linear combination of academic, behavioral, and demographic variables. Each coefficient represents the expected change in the final grade for a one-unit increase in that variable, while holding all other variables constant.

One of the most important predictors is G2, which has a coefficient of 0.8805. This means that for every one-point increase in a student's second period grade, their final grade is expected to increase by about 0.88 points, assuming everything else stays the same. This shows that recent performance in the course is a very strong indicator of how a student will finish.

Similarly, G1 has a coefficient of 0.1244, meaning that a one-point increase in the first period grade is associated with about a 0.12 point increase in the final grade. While its effect is smaller than G2, it still supports the idea that earlier performance in the course matters significantly.

The variable failures has a coefficient of -0.1976, which indicates that each additional prior course failure is associated with about a 0.20 point decrease in the final grade, holding all other

variables constant. This suggests that past academic struggles continue to impact student performance in the long run.

Model Quality and Usefulness

The model shows strong explanatory and predictive performance. The in-sample R^2 is approximately 0.858, meaning that about 85.8% of the variation in final grades is explained by the predictors in the model. When tested on new data, the model achieves an R^2 of 0.8499, which is very close to the training result and suggests that the model generalizes well. Additionally, standard regression diagnostics were considered, including checking for unusually large residuals and potential multicollinearity, to ensure model assumptions were reasonably satisfied.

The test set RMSE is approximately 1.0026, meaning that, on average, the model's predictions are off by about one grade point. This is a relatively small error and indicates that the model is fairly accurate in predicting student performance. The small gap between training and testing performance also suggests that the model is not heavily overfitting, even though it includes many predictors.

Overall, the model is very useful for predicting final grades and identifying key factors that influence student performance. Its strong performance is mainly driven by variables like G1 and G2, which reflect earlier performance in the same course. Because of this, the model works especially well for prediction, but some of its strongest predictors are based on prior outcomes rather than underlying behaviors or external factors. Therefore, this model is highly effective for predicting final grades and identifying the most important academic factors influencing student performance.

Appendix:

Code

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
# 1. LOAD DATA
df =
pd.read_csv("C:\\Users\\uemur\\Desktop\\DTSC\\DTSC2302\\student-performance_n_600.csv"
)
# Drop unnecessary index column if it exists
if "Unnamed: 0" in df.columns:
   df = df.drop(columns=["Unnamed: 0"])
# 2. DEFINE TARGET AND PREDICTORS
# Part I uses ALL predictors for G3
y = df["G3"]
X = df.drop(columns=["G3"])
# 3. ENCODE CATEGORICAL VARIABLES
X = pd.get_dummies(X, drop_first=True)
# 4. TRAIN / TEST SPLIT
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.20, random_state=2302)
# 5. ADD CONSTANT FOR OLS
X_train_sm = sm.add_constant(X_train)
X_test_sm = sm.add_constant(X_test, has_constant="add")
# 6. FIT MODEL ON TRAINING DATA
final_model = sm.OLS(y_train, X_train_sm.astype(float)).fit()
# 7. OUT-OF-SAMPLE PREDICTIONS
y_pred = final_model.predict(X_test_sm.astype(float))
# 8. EVALUATE OUT-OF-SAMPLE PERFORMANCE
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("Final Model Test RMSE:", round(rmse, 4))
print("Final Model Test R^2:", round(r2, 4))
print(final_model.params[["G1", "G2"]])
# 9. MODEL SUMMARY FOR APPENDIX
print(final_model.summary())
```

Model Summary

```
Final Model Test RMSE: 1.0026
Final Model Test R^2: 0.8499
G1    0.124438
G2    0.880535
dtype: float64
                            OLS Regression Results
==============================================================================
Dep. Variable:                     G3   R-squared:                       0.858
Model:                            OLS   Adj. R-squared:                  0.844
Method:                 Least Squares   F-statistic:                     64.37
Date:                Fri, 20 Mar 2026   Prob (F-statistic):          3.72e-159
Time:                        14:15:10   Log-Likelihood:                -795.12
No. Observations:                 480   AIC:                             1674.
Df Residuals:                     438   BIC:                             1850.
Df Model:                          41
Covariance Type:            nonrobust
==================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const                 0.8580      1.201      0.714      0.475      -1.503       3.219
age                   0.0315      0.060      0.526      0.599      -0.086       0.149
Medu                 -0.1202      0.090     -1.341      0.181      -0.296       0.056
Fedu                  0.0821      0.082      1.008      0.314      -0.078       0.242
traveltime            0.0988      0.092      1.072      0.284      -0.082       0.280
studytime             0.0450      0.085      0.527      0.598      -0.123       0.213
...
==============================================================================
```